

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.naive_bayes import CategoricalNB
from sklearn.metrics import accuracy_score, classification_report
```

```
data = pd.read_csv("mushrooms.csv")
print("First 5 rows:")
print(data.head())
print("\nColumn Names:")
print(data.columns)
print("\nDataset Shape:")
print(data.shape)
```

First 5 rows:

	class	cap-shape	cap-surface	cap-color	bruises	odor	gill-attachment	\
0	p	x	s	n	t	p		f
1	e	x	s	y	t	a		f
2	e	b	s	w	t	l		f
3	p	x	y	w	t	p		f
4	e	x	s	g	f	n		f

	gill-spacing	gill-size	gill-color	...	stalk-surface-below-ring	\
0	c	n	k	...		s
1	c	b	k	...		s
2	c	b	n	...		s
3	c	n	n	...		s
4	w	b	k	...		s

	stalk-color-above-ring	stalk-color-below-ring	veil-type	veil-color	\
0		w		p	w
1		w		p	w
2		w		p	w
3		w		p	w
4		w		p	w

	ring-number	ring-type	spore-print-color	population	habitat
0	o	p	k	s	u
1	o	p	n	n	g
2	o	p	n	n	m
3	o	p	k	s	u
4	o	e	n	a	g

[5 rows x 23 columns]

Column Names:

```
Index(['class', 'cap-shape', 'cap-surface', 'cap-color', 'bruises', 'odor',
      'gill-attachment', 'gill-spacing', 'gill-size', 'gill-color',
      'stalk-shape', 'stalk-root', 'stalk-surface-above-ring',
```

```

    'stalk-surface-below-ring', 'stalk-color-above-ring',
    'stalk-color-below-ring', 'veil-type', 'veil-color', 'ring-number',
    'ring-type', 'spore-print-color', 'population', 'habitat'],
    dtype='object')

```

Dataset Shape:
(8124, 23)

```

le = LabelEncoder()
for col in data.columns:
    data[col] = le.fit_transform(data[col])
print("\nAfter Encoding:")
print(data.head())
print("\nTarget Encoding: 0 = Edible, 1 = Poisonous")

```

After Encoding:

	class	cap-shape	cap-surface	cap-color	bruises	odor	gill-attachment
0	1	5	2	4	1	6	1
1	0	5	2	9	1	0	1
2	0	0	2	8	1	3	1
3	1	5	3	8	1	6	1
4	0	5	2	3	0	5	1

	gill-spacing	gill-size	gill-color	...	stalk-surface-below-ring	\
0	0	1	4	...	2	
1	0	0	4	...	2	
2	0	0	5	...	2	
3	0	1	5	...	2	
4	1	0	4	...	2	

	stalk-color-above-ring	stalk-color-below-ring	veil-type	veil-color	\
0	7	7	0	2	
1	7	7	0	2	
2	7	7	0	2	
3	7	7	0	2	
4	7	7	0	2	

	ring-number	ring-type	spore-print-color	population	habitat
0	1	4	2	3	5
1	1	4	3	2	1
2	1	4	3	2	3
3	1	4	2	3	5
4	1	0	3	0	1

[5 rows x 23 columns]

Target Encoding: 0 = Edible, 1 = Poisonous

```

X = data.drop('class', axis=1)
Y = data['class']
print("\nFeatures (X):")
print(X.head())

```

```

le = LabelEncoder()
for col in data.columns:
    data[col] = le.fit_transform(data[col])
print("\nAfter Encoding:")
print(data.head())
print("\nTarget Encoding: 0 = Edible, 1 = Poisonous")

```

Features (X):

	cap-shape	cap-surface	cap-color	bruises	odor	gill-attachment	\
0	5	2	4	1	6		1
1	5	2	9	1	0		1
2	0	2	8	1	3		1
3	5	3	8	1	6		1
4	5	2	3	0	5		1

	gill-spacing	gill-size	gill-color	stalk-shape	...	\
0	0	1	4	0	...	
1	0	0	4	0	...	
2	0	0	5	0	...	
3	0	1	5	0	...	
4	1	0	4	1	...	

	stalk-surface-below-ring	stalk-color-above-ring	stalk-color-below-ring	\
0		2	7	7
1		2	7	7
2		2	7	7
3		2	7	7
4		2	7	7

	veil-type	veil-color	ring-number	ring-type	spore-print-color	\
0	0	2	1	4		2
1	0	2	1	4		3
2	0	2	1	4		3
3	0	2	1	4		2
4	0	2	1	0		3

	population	habitat
0	3	5
1	2	1
2	2	3
3	3	5
4	0	1

[5 rows x 22 columns]

After Encoding:

	class	cap-shape	cap-surface	cap-color	bruises	odor	gill-attachment	\
0	1	5	2	4	1	6		1
1	0	5	2	9	1	0		1
2	0	0	2	8	1	3		1
3	1	5	3	8	1	6		1

4	0	5	2	3	0	5	1
---	---	---	---	---	---	---	---

	gill-spacing	gill-size	gill-color	...	stalk-surface-below-ring	\
0	0	1	4	...		2
1	0	0	4	...		2
2	0	0	5	...		2
3	0	1	5	...		2
4	1	0	4	...		2

	stalk-color-above-ring	stalk-color-below-ring	veil-type	veil-color	\
0		7	7	0	2
1		7	7	0	2
2		7	7	0	2
3		7	7	0	2
4		7	7	0	2

	ring-number	ring-type	spore-print-color	population	habitat
0	1	4	2	3	5
1	1	4	3	2	1
2	1	4	3	2	3
3	1	4	2	3	5
4	1	0	3	0	1

[5 rows x 23 columns]

Target Encoding: 0 = Edible, 1 = Poisonous

```
X_train, X_test, Y_train, Y_test = train_test_split(
    X, Y, test_size=0.2, random_state=42
)
print("\nTraining Data Shape:", X_train.shape)
print("Testing Data Shape:", X_test.shape)
```

Training Data Shape: (6499, 22)
Testing Data Shape: (1625, 22)

```
nb_model = CategoricalNB()
nb_model.fit(X_train, Y_train)
print("\nNaive Bayes Model Trained Successfully")
```

Naive Bayes Model Trained Successfully

```
y_pred = nb_model.predict(X_test)
print("\nFirst 10 Predictions:")
print(y_pred[:10])
```

First 10 Predictions:
[0 1 1 0 1 1 1 1 0 0]

```
accuracy = accuracy_score(Y_test, y_pred)
print("\nAccuracy:", accuracy)
```

```
print("\nClassification Report:")
print(classification_report(Y_test, y_pred))
```

Accuracy: 0.9507692307692308

Classification Report:

	precision	recall	f1-score	support
0	0.92	0.99	0.95	843
1	0.99	0.91	0.95	782
accuracy			0.95	1625
macro avg	0.96	0.95	0.95	1625
weighted avg	0.95	0.95	0.95	1625

```
new_data = X.iloc[0:1]
print("\nNew Data Sample:")
print(new_data)
prediction = nb_model.predict(new_data)
print("\nNew Data Prediction (0=Edible, 1=Poisonous):", prediction[0])
```

New Data Sample:

	cap-shape	cap-surface	cap-color	bruises	odor	gill-attachment	\
0	5	2	4	1	6		1
	gill-spacing	gill-size	gill-color	stalk-shape	...		\
0	0	1	4	0	...		
	stalk-surface-below-ring	stalk-color-above-ring	stalk-color-below-ring				
0		2		7			7
	veil-type	veil-color	ring-number	ring-type	spore-print-color		\
0	0	2	1	4		2	
	population	habitat					
0	3	5					

[1 rows x 22 columns]

New Data Prediction (0=Edible, 1=Poisonous): 0